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Creating a corpus of plagiarised academic texts

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Abstract

Plagiarism is a serious problem in higher education and generally acknowledged to be on the increase (McCabe, 2005). Text analysis tools have the potential to be applied to work submitted by students and assist the educator in the detection of plagiarised text. It is difficult to develop and evaluate such systems without examples of such documents. There is therefore the need for resources that contain examples of plagiarised text submitted by students. However, gathering examples of such texts presents a unique set of challenges for corpus construction.

This paper discusses current work towards the creation of a corpus of documents submitted for assessment in higher education that contain examples of simulated plagiarism. The corpus is designed to represent the types of plagiarism that are found within higher education as closely as possible. We describe the process of corpus creation and some features of the resulting resource. It is hoped that this resource will become useful for research into the problem of plagiarism detection.

1 Introduction

In recent years plagiarism (and its detection) has received much attention within the academic and commercial communities (e.g. (Hislop, 1998; Joy, 1999; Lyon et. al., 2001; Colberg and Kobourov, 2005; Eissen and Stein, 2006; Kang et. al., 2006). In academia students have used technology to fabricate texts (e.g. using pre-written texts from essay banks or paper mills, using word processors to manipulate texts and finding potential source texts using online search engines) and plagiarism is now widely acknowledged to be a significant and increasing problem for higher education institutions (Colwin and Lancaster, 2001; Zobel, 2004; McCabe, 2005).

The academic community have suggested a wide range of approaches to the detection of plagiarism, for example (White and Joy, 2004; Colberg and Kobourov, 2005), and many commercial systems are also available (Bull, 2001). However, one of the barriers preventing a comparison between these techniques is the lack of a standardised evaluation resource. Such a resource would enable a quantitative evaluation of existing techniques for plagiarism detection. Standardised evaluation resources have been very beneficial to a wide range of fields including Information

Retrieval (Voorhees and Harman, 2005), Natural Language Processing (Grishman and Sundheim, 1996; Mihalcea et. al., 2004) and authorship attribution (Juola, 2006).

Unfortunately the process of creating a suitable corpus is not straightforward. Firstly, there are a variety of types of plagiarism (see Section 2) and it may not be practical to include them all in a single resource. In addition, the collection of plagiarised documents raises challenges that are not present in the majority of corpus construction exercises. Firstly, plagiarism is essentially an act of deception. A student who plagiarises does not intend for the plagiarism to be discovered and may also be unlikely to admit that a text is plagiarised. Consequently it may not be possible to identify the documents that we aim to include in a plagiarism corpus. In addition, even if it were possible to identify plagiarised documents, it is unlikely that they could be made freely available for research purposes. The document's writer is unlikely to agree to this and doing so is likely to be regarded as ethically, and perhaps also legally, unacceptable. These issues form a significant challenge to any attempt to create a benchmark corpus of plagiarised documents.

Suggestions have been made for automatically created plagiarism corpora (see Section 2), however these are limited in various ways. This paper describes the construction of a plagiarism corpus. To avoid the problems involved in collecting genuine examples of plagiarism we chose to simulate plagiarism by asking authors to intentionally reuse another document in a way that would normally be regarded as unacceptable (see Section 3). The corpus is not intended to comprehensively represent all possible types of plagiarism but does contain types that are not included in the resources that are currently available (see Section 2). The corpus is analysed to gain insight into the strategies used by students (Section 4). It is suggested that this corpus forms a valuable addition to the set of available resources for the plagiarism detection task.

2 Background

2.1 Varieties of Plagiarism Analysis

A range of problems has been explored within the study of plagiarism analysis. Stein (2006) distinguishes *extrinsic* and *intrinsic* plagiarism analysis. In the first case the aim is to identify plagiarised portions of text *within* documents and the corresponding source; whilst the second case describes the scenario where the source does not need to be identified.

In extrinsic plagiarism analysis a key factor is the comparison of portions of text that it is suspected are plagiarised with their potential sources. This problem is made complex by the fact that there are a wide variety of "levels" of plagiarism. Martin (1994) points out that these include word-for-word plagiarism (direct copying of phrases or passages from another text without quotation or acknowledgment), paraphrasing plagiarism (when words or syntax are rewritten, but the source text can still be recognised) and plagiarism of ideas (the reuse of an original idea from a source text without dependence on the words or form of the source). Meyer zu Eissen et. al. (2007) and Pinto et. al. (2009) also point out that the source could be written in

a different language and have been translated (either automatically or manually) before being reused.

The problem, however, is a different one in the case of intrinsic plagiarism analysis. In this case the aim is to identify portions of text that are somehow distinct from the rest of the document, for example a significant improvement in grammar or discussion of more advanced concepts than would be expected, and might raise suspicion in a human reader.

There may also be variation in the number of source texts that have been plagiarised. A document may plagiarise a single source; the most extreme version of this situation is when an original document is copied verbatim and the author changed (Martin, 1994). Plagiarism of this type may also include modifications to the original document or a plagiarised section being included as part of an otherwise acceptable document. Alternatively, a document may plagiarise from more than one source and, similarly, the document may consist only of plagiarised passages or plagiarised sections embedded within it and these passages may be modified or used verbatim.

2.2 Existing Corpora

In order to evaluate approaches to plagiarism detection it is useful to have access to a corpus containing examples of the types of plagiarism that we aim to identify. Given the difficulties involved in obtaining examples of plagiarised texts an attractive approach is to develop a corpus automatically. For example, Meyer zu Eissen et. al. (2007) created a corpus for plagiarism detection experiments by manually adapting Computer Science articles from the ACM digital library (Web Technology & Information Systems Group, 2008) by adding passages from other articles to simulate plagiarism. Some of these passages were copied verbatim while others were altered. However, Meyer zu Eissen et. al. (2007) do not describe the process of corpus creation in detail. A corpus was also automatically created for the 2009 PAN Plagiarism Detection Competition[1]. This resource contains texts of a wide range of lengths containing differing amounts of texts inserted from other documents. The reused text is either obfuscated, by randomly moving words or replacing them with a related lexical item, or translated from a Spanish or German source document (Potthast et al., 2009). Guthrie et. al. (2007) also simulated plagiarism by inserting a section of text written by another author into a document, although they did not alter the inserted text in any way.

This approach is convenient since it allows corpora of "plagiarised" documents to be created with little effort. In fact, if the inserted passages are not altered, as Guthrie et. al. chose to do, the amount of documents that could be created are only limited by the size of the collection. However, it is not clear the extent to which these corpora reflect the types of plagiarism that might be encountered in academic settings.

While plagiarism is an unacceptable form of text re-use there are other forms of this practice that are not objectionable, such as the reuse of news agency text by newspapers. The METER Corpus[2] is a hand-crafted collection of 1,716 texts built

[1] http://www.webis.de/pan-09
[2] http://www.dcs.shef.ac.uk/nlp/meter/Metercorpus/metercorpus.htm

specifically for the study of text reuse between newswire source texts and stories published in a range of British national newspapers (Clough et. al., 2002). The corpus contains a collection of news stories between July 1999 and June 2000 in two domains: (1) law and court reporting, and (2) show business and entertainment. The newspaper articles were analysed to identify the degree to which they were derived from the news agency source and annotated with a three level scheme that indicated whether the text was entirely, partially or not derived from the agency source. Almost half of the stories were analysed in more detail to identify whether the text was extracted verbatim from the news agency text, rewritten or completely new. The METER corpus is freely available and contains detailed annotation at a level which could be very valuable in the development of plagiarism detection systems, however, the main drawback of this corpus is that the type of text reuse it represents is not plagiarism.

Plagiarism may involve attempts to disguise the source text and this may be attempted by paraphrasing (see Section 3.2 for further discussion). Within the field of Computational Linguistics there as been interest in the identification and generation of paraphrases over the last decade, for example (Barzilay and McKeown, 2001; Callison-Burch et. al., 2006). This has lead to the development of a variety of corpora containing examples of paraphrases and, while these do not represent text reuse, they are potentially valuable for evaluating some aspects of plagiarism detection. Example paraphrase corpora include, the Microsoft Research Paraphrase Corpus[3] (MSRPC, see (Dolan et. al., 2004)) contains almost 6,000 pairs of sentences obtained from Web news sources that have been manually labelled to indicate whether the two sentences are paraphrases or not. The Multiple-Translation Chinese Corpus[4] (MTCC, see (Pang et. al., 2003)) makes use of the fact that translators may choose different phrases when translating the same text. The corpus consists of 11 independent translations of 993 sentences of journalistic Mandarin Chinese text. Cohn et. al. (2008) recently described a corpus[5] consisting of parallel texts in which paraphrases were manually annotated. While these resources are potentially useful in the development of plagiarism detection systems they are limited by the fact that, like the METER corpus, they consist of acceptable forms of text reuse.

The various corpora relevant to the plagiarism detection are limited since there is no guarantee that they represent the types of plagiarism that may be observed in practice. Artificially created corpora are attractive, and allow data sets to be created quickly and efficiently, but may be limited to one type of plagiarism (insertion of reused section in an otherwise valid document) and, if the inserted text is altered, it may not be changed in the same way a student may choose to. In addition, the various resources based on acceptable forms of text reuse (including the METER corpus and paraphrase corpora) do not include the element of deception involved in plagiarism.

3. Corpus Creation

[3] http://research.microsoft.com/en-us/downloads/607D14D9-20CD-47E3-85BC-A2F65CD28042/default.aspx
[4] http://www.ldc.upenn.edu/Catalog/CatalogEntry.jsp?catalogId=LDC2002T01
[5] http://homepages.inf.ed.ac.uk/tcohn/paraphrase_corpus.html

We aim to create a corpus that could be used for the development and evaluation of plagiarism detection systems that reflects the types of plagiarism practiced by students in an academic setting as far as realistically possible. We decided to avoid the strategies used in the creation of related corpora (see Section 2.2) since these may not accurately represent these types of plagiarism. We did not have the resources available to create a resource that includes all possible types of plagiarism (see Section 2.1) and decided to focus instead on examining a variety of rewrite levels in the scenario where a single source is plagiarised.

The strategy we adopted was to create a set of exercises ("learning tasks") that undergraduate students might be asked to complete (see Section 3.1). A set of participants were recruited (see Section 3.3) and asked to complete these exercises using a variety of approaches designed to simulate situations in which the exercises were completed by plagiarising another text and without any plagiarism (see Section 3.2).

3.1 Learning Tasks

We created a set of five short answer questions on a variety of topics that might be found in an undergraduate Computer Science curriculum. Short answer questions were chosen since they provide an opportunity for plagiarism at the same time as minimising the burden placed on participants.

1. What is inheritance in object oriented programming?
2. Explain the PageRank algorithm that is used by the Google search engine.
3. Explain the Vector Space Model that is used for Information Retrieval.
4. Explain Bayes Theorem from probability theory.
5. What is dynamic programming?

This set of questions were chosen to represent a range of areas of Computer Science and also designed to be such that it was unlikely for any student to know the answer to all five questions. In addition, materials that are necessary for participants to answer these questions (see Section 3.2) could be easily obtained and provided to participants. The questions can essentially be answered by providing a short definition of the concept being asked about. Although some of the questions allow for relatively open-ended discussion they can be adequately answered using a few hundred words.

3.2 Generation of Answers

For each of these questions we aim to create a set of answers using a variety of approaches, some of which simulate cases in which the answer is plagiarised and others that simulate the case in which it is not plagiarised. To simulate plagiarism we require a source text in which the answer is found and used Wikipedia[6] for this. Wikipedia was chosen since it is readily available, generally accepted to provide information on a wide variety of topics and contains versions of pages in multiple

[6] http://www.wikipedia.com

languages (thus allowing evaluation of cross-lingual plagiarism detection) and contained answers to the type of questions used in our study.

We aimed to represent a variety of different degrees of rewrite in the plagiarised documents to enable the evaluation of different plagiarism detection algorithms. This is similar to proposals for levels of plagiarism in software code (Faidhi and Robinson, 1987) adapted for texts. Keck (2006) discusses the following "levels" of rewrite: Near Copy, Minimal Revision, Moderate Revision, and Substantial Revision. These represent progressively more complex (and difficult) forms of rewrite identified from a set of plagiarised examples. Rewriting operations resulting from plagiarism may involve verbatim cut and paste, paraphrasing and summarising (Keck, 2008). Cut and paste involves lifting the original text with only very minor, if any, changes. Paraphrases are alternative ways of conveying the same information (Barzilay and McKeown, 2001) using lexical items or syntax. Campbell (1990) and Johns and Myers (1990) suggest that paraphrasing is one of a number of strategies (including summary and quotation) that students can use when integrating source texts into their writing. A summary is (typically) a shortened version of an original text. A summary should include all main ideas and important details, while reflecting the structure and order of the original. Editing operations typically used in producing summaries include splitting up sentences from the original (sentence reduction), combining multiple sentences from the original (sentence combination), syntactic transformations (paraphrasing), lexical paraphrasing, the generalisation or the specification of concepts in the original text, and the reordering of sentences (Jing and McKeown, 1999).

To generate our corpus, participants were asked to answer each question using one of four methods:

- **Near copy** Participants were asked to answer the question by simply copying text from the relevant Wikipedia article (i.e. performing cut-and-paste actions). No instructions were given about which parts of the article to copy (selection had to be performed to produce a short answer of the required length of between 200-300 words).

- **Light revision** Participants were asked to base their answer on text found in the Wikipedia article and participant were, once again, given no instructions about which parts of the article to copy. They were instructed that they could alter the text in some basic ways including substituting words and phrases with synonyms and altering the grammatical structure (i.e. paraphrasing). Participants were also instructed not to radically alter the order of information found in sentences.

- **Heavy revision** Participants were once again asked to base their answer on the relevant Wikipedia article but were instructed to rephrase the text to generate an answer with the same meaning as the source text, but expressed using different words and structure. This could include splitting source sentences into one or more individual sentences, or combining more than one source sentence into a single sentence. No constraints were placed on how the text could be altered.

- **Non-plagiarism** Participants were provided with learning materials in the form of either lecture notes or sections from textbooks that could be used to answer the relevant question. Participants were asked to read these materials and then attempt to answer the question using their own knowledge (including what they had just learned from the materials provided). They were also told that they could look at other materials to answer the question but explicitly instructed not to look at Wikipedia.

The aim of the final method (non-plagiarism) was to simulate the situation in which a student is taught a particular subject and their knowledge subsequently tested in some form of assessment. It is important to remember that just because a student has been taught a particular topic does not necessarily mean that they will be able to answer questions about it correctly and that one of the functions of assessment is to determine whether or not a student has mastered material they have been taught. The non-plagiarism scenario was included since it is useful to determine whether it is possible to distinguish between answers that are intentionally plagiarised and those where the student has attempted to understand the question before answering. Non-plagiarised answers also indicate the amount of text that is likely to be shared by independently written answers to the same questions.

3.3 Participation

A total of 19 participants were recruited to create texts for the corpus. Five were members of a team carrying out a group project on plagiarism detection while the remaining participants were either recruited by this group (through personal contact) or responded to an email request for volunteers. All participants were students in the Computer Science Department of Sheffield University and were studying for a degree in Computer Science at either undergraduate or postgraduate level. Participation was restricted to students with some familiarity of Computer Science since some familiarity with the topic would be required to answer the questions and also that this provided a more realistic plagiarism scenario.

Participants were presented with each of the five questions and asked to provide a single answer to each. They were instructed that answers should be between 200 and 300 words long and, to simplify later processing, should contain only standard (ASCII) characters and avoid using any symbols, graphics or computer code. For each question participants were instructed which approach to use to provide the answer. Two of the five questions were answered without plagiarising (the “non-plagiarism” category), one question using the near copy category, one using light revision and one using heavy revision. The approach used for each question varied between participants to provide a variety of different answers to each question. To reduce learning and order effects, the tasks and categories used were arranged using a Graeco-Latin square arrangement. An alternative methodology would have been to ask a single participant to provide multiple answers to each question, using a variety of approaches, however, this could have caused problems since the process of answering a question using one approach could influence subsequent answers.

All participants provided written consent to allow us to use their answers in order to make the corpus publicly accessible. Participants were also asked to complete a short

questionnaire after answering the questions. This recorded whether or not they were a native English speaker and, for each question, how familiar they were with the answer to the question being asked and how difficult they found answering the question. Finally, participants were provided with a small reward for participation (electronic voucher for an on-line store).

4. Corpus Analysis

4.1 Corpus Profile

The corpus contains 100 documents (95 answers provided by the 19 participants and the five Wikipedia source articles). The answer texts contain 19,559 words in total (2,2230 unique tokens). The Wikipedia pages total 14,242 words after conversion to plaintext and removal of URL references. The average length of file in the corpus is 208 words (std dev. 64.91) and 113 unique tokens (std dev. 30.11). Native English speakers created 59 (62%). The first language varied for the authors of the remaining texts. For each learning task there are 19 examples of each of the heavy revision, light revision and near copy levels and 38 non-plagiarised examples written independently from the Wikipedia source.

4.2 Observations

The corpus has some interesting features that are unlikely to be found in other resources. Unlike the majority of corpora, which consist of carefully edited texts, our corpus includes documents that contain spelling, grammatical and typographical errors. Figure 1 shows extracts from answers provided for learning task A (see Section 3.2) containing such errors. (The spacing in these examples is as provided by the participants.) It should be noted, however, these extracts represent two of the more extreme examples of errors found within texts. No attempt was made to clean up the texts to remove these errors from the texts since doing so would alter the material provided by the participants and these errors may actually complicate the task of plagiarism detection (by hampering string overlap approaches and making deeper analysis more difficult).

Example 1: Inheritance allowes classes to be categorized, similer to the way humans catagorize. It also provides a way to generalize du to the ``is a'' relationship between classes.

Example 2: Generlisation also some time known as inheritance. The main reason behind this is a hierarchi st ructure of objects and classes. We can understand this mechanism by some examples: like fruit i s aq main class and mangoes apple ,orange is child classs of the main class.So obviously inherit all the properties of fruit class.

Figure 1: Examples of answers to Learning Task A containing errors

The simplest type of rewrite included in our study was cut-and-paste (near copy). Although this option did not require the participant to alter the text, they still had to

decide which parts of the relevant Wikipedia article to use in their answer since the articles were longer than the 200-300 words requested. Participants used a variety of strategies including simply copying a single contiguous sequence of text of roughly the required length; others selected the portions of the text that most directly answered the relevant question. This could involve deleting isolated sentences or choosing sentences from throughout the article that are recombined into a coherent answer.

When participants were asked to perform light or heavy revision they employed similar strategies for selecting portions of the text from the Wikipedia source. Figure 2 shows examples of light and heavily revised sentences and the corresponding sentence in the Wikipedia source for Learning Tasks A. In the examples of light revision the connection between the source and plagiarised text is generally obvious (at least to the human). A number of techniques were used to obscure the connection with the source text. The first example of a lightly revised response demonstrates deletion (the phrases "In object-oriented programming," and "(instances of which are called objects)" are removed), substitution of words with synonyms ("way" becomes "method") and simple paraphrases ("to form" becomes "of forming" and "classes that have already been defined" becomes "predefined classes"). A common strategy in the examples of heavy revision is to obscure the link to the source text further by altering the amount of information contained in each sentence, either to include something from an additional sentence or to break single sentence into two separate sentences. For example, in Figure 2, in the first example of heavy revision contains information from the sentence immediately following the source. The second example includes information from various parts of the source article.

Wikipedia source sentence: In object-oriented programming, inheritance is a way to form new classes (instances of which are called objects) using classes that have already been defined.
Light revision example 1: Inheritance is a method of forming new classes using predefined classes.
Light revision example 2: The idea of inheritance in OOP refers to the formation of new classes with the already existing classes.
Heavy revision example 1: When we talk about inheritance in object-oriented programming languages, which is a concept that was invented in 1967 for Simula, we are usually talking about a way to form new classes and classes are instances of which are called objects and involve using classes that have already been defined.
Heavy revision example 2: Object oriented programming is a style of programming that supports encapsulation, inheritance, and polymorphism. Inheritance means derived a new class from the base class. We can also say there are parents class and child classes in inheritance.

Figure 2: Examples of light and heavily revised sentences in answers to Learning Task A

5 Summary and Future Work

In this paper we have discussed the creation of a corpus designed to assist in the evaluation of plagiarism detection systems for (English) natural language texts. Our aim was to generate a resource that represented the strategies used by students when reusing text as far as is possible. Rather than relying on automatic methods for generating plagiarised texts our resource consists of examples manually generated by students at our institution. The importance of generating realistic examples has been highlighted through the analysis of plagiarised texts where aspects such as language skills have demonstrated that the examples may contain a range of grammatical, typographical and spelling errors. We hope that this corpus will (1) enable comparative evaluation between existing and new techniques for automated plagiarism detection, (2) help stimulate further research in the field, (3) help us to understand the strategies used by students when they plagiarise, and (4) be of potential use as a pedagogical resource to provide examples of plagiarism.

The corpus is a work in progress and is limited in a number of ways. Some of these shortcomings, and plans for future work to address them, are outlined below.

- **Corpus size**. It is likely that the set of techniques that could be applied will be limited by lack of data currently available. It would be possible to increase the number of texts by repeating the corpus construction process described in this paper since the necessary resources (learning tasks, Wikipedia article and material for non-plagiarised answer) are readily available. However, our approach is resource-intensive which may prove a limiting factor on the size of corpus that can be generated.

- **Academic discipline**. The corpus is limited to the field of Computer Science. The main reason for this decision was that we could easily create the necessary resources ourselves and had unhindered access to participants. However, we do not know whether the strategies applied when reusing text vary for students studying different academic disciplines and we plan to explore others in future work. The process of corpus creation outlined here could be adapted to other disciplines in a straightforward way by creating learning tasks and identifying suitable material. A suitably qualified person in the relevant academic discipline should create these resources. We are interested in examining as wide a range of disciplines as possible.

- **Learning task**. The exercise completed by our participants, short answer questions, represents only one type of assessment that is used within higher education. Short answer questions are convenient since they can be completed relatively quickly and allowed us to ask participants to answer multiple questions. However, the amount of text generated for each question is limited to just a few hundred words and only represents one possible assessment method. For future work we intend to explore learning tasks that require longer answers since this will allow more text to be generated and allow the inclusion of more varied types of plagiarism.

- **Types of plagiarism**. Participants were asked to plagiarise from a single source (a Wikipedia article) and, while a variety of types of rewrite were used,

this represents just one type of plagiarism (see Section 2.1). We plan to explore further types of plagiarism including providing participants with more than one source document that can be plagiarised. We also plan to explore whether it might be practical to provide participants with source text in another language that they could translate (either manually or automatically).

- **Annotation**. In the current version of the corpus each document is labelled with the method the participant was asked to use when creating it (i.e. near copy, light revision, heavy revision or non-plagiarism). Annotating the corpus at a more fine-grained level is likely to provide extremely valuable information. For example, it may be possible to learn the rewrite strategies that are applied if plagiarised text in a document could be linked to the relevant source text in Wikipedia. However, the extent to which this could be achieved is open to question and experience from a similar annotation exercise on a corpus of news stories (Clough et. al., 2002) suggest that this is likely to be an extremely time consuming process.

- **Analysis of plagiarist behaviour**. The process used to create our corpus is unrealistic since the participants were given guidance about rewriting text and provided with a source to reuse. This was a useful way to generate a variety of documents for each learning task and the types of rewrite were based on previous work on plagiarism in higher education (see Section 2.1). However, our ultimate aim is to create a corpus of documents that represents the types of plagiarism actually used in practise as far as possible. It would be interesting to explore the behaviour of plagiarists in more detail and determine how closely the process applied here simulated this. Ideally we would like to create a set of controlled conditions that resemble the true plagiarism scenario as closely as possible. Participants would attempt to deceive by making unfair use of a source text without being instructed to do so, or even being aware of the purpose of the exercise, while trying to obfuscate the reuse. It is, however, open to question whether it is possible to create these conditions in a useful way.

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